Investigation: METALURGICA DUQUE SA (CVM code 8648), fiscal year 2011, as it was on 2013-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2011 DFP, version 1, filed on 2012-03-30; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2012-03-30).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=15754

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −1.866 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 91 (n = 17)
- Percentile in group and year: 82 (n = 363)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.908 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 93 (n = 23)
- Percentile in group and year: 93 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 0.862 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 2 (n = 23)
- Percentile in group and year: 17 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.816 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 20 (n = 23)
- Percentile in group and year: 12 (n = 462)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.065 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 59 (n = 23)

Percentile in group and year: 38 (n = 468)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.059 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 62 (n = 17)
Percentile in group and year: 71 (n = 377)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 1.002 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 46 (n = 23)
Percentile in group and year: 47 (n = 461)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.013 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 54 (n = 23)
Percentile in group and year: 63 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.242 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 85 (n = 23)
Percentile in group and year: 84 (n = 476)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 28.0 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 34.2 days)
Percentile in sector and year: 93 (n = 22)
Percentile in group and year: 88 (n = 441)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 11.87 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 63 (n = 23)
Percentile in group and year: 67 (n = 429)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.014 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 59 (n = 23)
Percentile in group and year: 63 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 0.779 (index)
Components: X1 = −0.127; X2 = 0.061; X3 = 0.063; X4 = 0.943 (indices)
Percentile in sector and year: 27 (n = 24)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 7/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 1
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 1
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 1
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 1 accounts above the R$1,000 tolerance (group a, totals and key lines: 0; group b, reclassifications: 1), in the annex at the end.

What was known on the date

- DFP 2010, v1: 2011-03-22
- FCA 2011, v1: 2011-03-22
- FCA 2011, v2: 2011-06-03

- DFP 2011, v1: 2012-03-30
- DFP 2012, v1: 2013-03-28
DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality
53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)
Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2010 FCA: Audiplan Auditores Independentes.
Auditor in the 2011 FCA: Audiplan Auditores Independentes.
Auditor in the 2012 FCA: Audiplan Auditores Independentes.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested
In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits
- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations
- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine
motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E).
Generated at 2026-11-10T19:31:23+00:00.

Today's information (outside the reference date)
Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(b) DRE 3.99.01.01 ON: 0.0 in the version used, 0.0 in the comparative (difference of −0.0)